

Importance of root growth in overcoming planting stress

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Abstract. Root growth is critical to the establishment of planted seedlings. Seedlings can undergo stress just after planting if root growth is not sufficient to couple the seedling to available soil water. Stress occurs when a newly planted seedling's root system can not supply enough water to transpiring needles to maintain a proper water balance and ensure survival. Thus, a newly planted seedling's ability to overcome planting stress is affected by its root system size and distribution, root–soil contact, and root hydraulic conductivity. This paper describes how factors of root growth and water status of newly planted seedlings are important in overcoming the phenomenon of planting stress which then allows a newly planted seedling to enter the establishment phase of development.

Introduction

Root growth and seedling establishment are tied together whenever foresters talk about successful reforestation programs. Seedling establishment depends on site environmental conditions and seedling quality at time of planting (Rietveld 1989; Burdett 1990). These two factors come into play whenever a seedling is planted. How a seedling responds to the environment and starts to develop roots out into the surrounding soil determines whether a seedling survives the planting process.

A newly planted seedling can be coupled to the reforestation site only if it has access to available soil water to meet the atmospheric demand for water (Figure 1). This coupling is important because reforestation sites can present extreme environmental conditions that alter site heat exchange processes and soil water relations (Miller 1983). The ability of a seedling to take up water is affected by its root system size and distribution, root–soil contact, and root hydraulic conductivity. The seedling shoot system, which is exposed to the atmospheric demand for water, has transpirational water loss from needles which is determined by the degree of stomatal opening and needle area. Typically, newly planted seedlings have restricted root placement, low root system permeability and/or poor root–soil contact, which can limit water uptake from

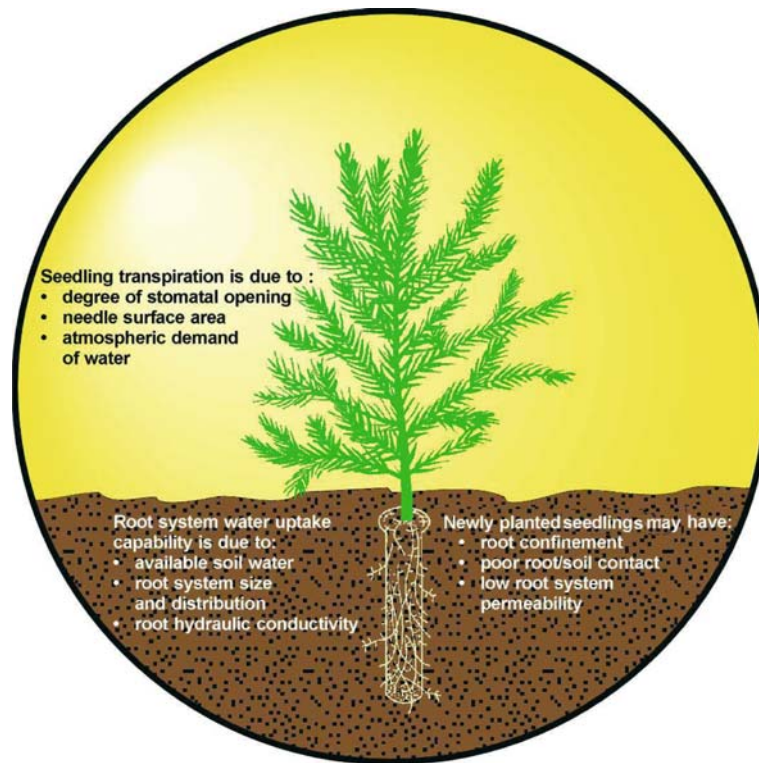


Figure 1. Descriptive representation of planting stress in a seedling (adapted from Grossnickle 2000).

the soil (Kozlowski and Davies 1975; Rietveld 1989; Burdett 1990). As a result, seedlings can be exposed to stress just after planting (i.e., planting stress) because they are not fully coupled into the hydrologic cycle whereby water flows from the soil to plant roots, through the plant and into the atmosphere. This planting stress can lead to root growth being limited by the lack of water and photosynthates, and in turn photosynthesis being limited by water stress due to a lack of root growth (Burdett 1990; Grossnickle 2000).

Seedlings that develop a root system after planting establish a proper water balance and respond to field site atmospheric conditions without limitations that occur when seedlings do not have access to soil water (Margolis and Brand 1990). Thus, seedlings with a favorable water status can have a cycle of root growth supported by photosynthesis, and photosynthesis supported by root growth (Burdett 1990). Seedlings enter the establishment phase when they are fully coupled into the site hydrological cycle and begin to respond to silvicultural practices that have been used to create favorable site conditions (Rietveld 1989; Grossnickle 2000). This paper discusses factors of seedling root

growth and water status, and their importance in overcoming the phenomenon of planting stress. This paper also provides examples of how planting stress can carry over into the establishment phase of development even after newly planted seedlings have developed a root system that couples them to available soil water.

Discussion

Root growth and seedling survival

It has long been known that new root growth during the first season in the field following planting is extremely important for seedling survival (Wakeley 1954; Stone 1955). This is because water stress can occur in newly planted seedlings and can result in mortality or limited early growth (Kozlowski and Davies 1975). In a number of surveys evaluating reasons for poor plantation establishment it has been determined that the greatest mortality on some sites occurs from planting stress (Vyse 1981; Waters et al. 1991). Thus, root growth after planting is critical for ensuring seedling establishment.

Recently planted seedlings initially depend on the root system created by nursery culture. Thus, during the period just after planting the original root system size determines the ability of seedlings to take up water so they can initiate the establishment process (Carlson and Miller 1990). Seedlings with larger root volumes have a better ability to take up water (Carlson 1986) avoid planting stress and become established after planting (McGilvray and Barnett 1982; Larsen et al. 1988; Blake et al. 1989; Rose et al. 1991a, b; Haase and Rose 1993). Therefore, seedling initial root system size in relation to shoot size can have a direct effect on seedling water balance just after planting (Burdett 1990; Folk and Grossnickle 1997). However, root system size alone will not avoid the planting stress phenomenon.

Trials have shown that new root growth is important for seedling survival. This phenomenon has been reported for a number of commercially planted conifer species [e.g., Stone 1955: Jeffrey pine (*Pinus jeffreyi* Grev. and Bal.), ponderosa pine (*Pinus ponderosa* Laws), white fir (*Abies concolor* Gord. & Glend. Lindl.); Burdett et al. 1984: white spruce (*Picea glauca* (Moench.) Voss.), lodgepole pine (*Pinus contorta* Dougl.); Hines and Long 1986: Engelmann spruce (*Picea engelmannii* Parry ex Engelm.); Larsen et al. 1988: loblolly pine (*Pinus taeda* L.); Simpson 1990: lodgepole pine, Douglas-fir (*Pseudotsuga menziesii glauca* (Mayr) Sudw), western hemlock (*Tsuga heterophylla* (Raf.) Sarg.)]. A good example is work conducted on interior spruce (*Picea glauca* (Moench.) Voss. × *Picea engelmannii* Parry ex Engelm.) seedlings (Figure 2). In this trial, survival was high when seedlings had the capability to grow more than 25–30 new roots, while survival declined to less than 35% in populations that had the capability to grow just a few or no new roots. These trials recognize the importance of new root growth in overcoming planting stress. Past

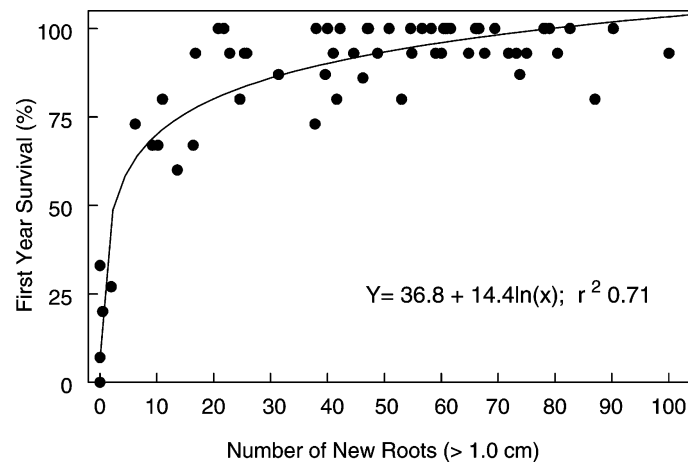


Figure 2. The relationship between first year survival on a reforestation site and the capability of interior spruce seedlings to grow roots at time of planting. Each point represents mean survival of 15 seedlings and mean root growth capacity of five seedlings taken from the same sample population (adapted from Simpson 1990).

synthesis of work on this topic has found ‘a compelling body of evidence’ showing a relationship between the ability of a seedling to grow roots and seedling survival after planting (Ritchie and Dunlap 1980). Simpson and Ritchie (1997) emphasized the importance of new root growth with increasing harshness of planting site conditions. Survival of newly planted seedlings improves when seedlings can extend new roots into the soil, ensuring that water intake equals or exceeds water loss (Wakeley 1954).

Any discussion on planting stress should recognize that stress is overcome only if seedlings have the proper combination of physiological and morphological attributes, and root growth capability to respond to reforestation site conditions. Seedlings need to have an adequate level of stress resistance to respond to drought and low temperature events (Ritchie 1984; Burdett 1990; Grossnickle and Folk 1993). Proper shoot to root balance is an important morphological attribute because it is a measure of seedling water loss and water uptake capability at the time of planting (Ritchie 1984; Thompson 1985; Burdett 1990; Mexal and Landis 1990; Grossnickle 2000). They note that seedlings with a proper shoot to root balance have the best potential for avoiding planting stress. Each tree species has an inherent seasonal root growth periodicity that is tied to dormancy related processes (Fuchigami and Nee 1987) and their seasonal cycle of morphological and physiological changes (Burr 1990; Ritchie and Tanaka 1990). Due to this periodicity of root growth, seedlings need to be planted at that point in their seasonal cycle when they have the ability to grow roots. Seedlings need to have all of the above attributes of a quality seedling to ensure the best chance of field establishment.

Root growth & reforestation site environment

Soil water, soil temperature, and the atmospheric demand for water are key environmental variables that determine whether seedlings are exposed to planting stress. The following is a brief discussion of how these variables can affect the growth response of newly planted seedlings. Discussions on the microclimate around field planted seedlings can be found elsewhere (e.g., Gates 1980; Geiger 1980; Spittlehouse and Stathers 1990; Keenan and Kimmins 1993; Grossnickle 2000).

Soil water

Soil water content and hydraulic properties of the soil system can affect whether stress occurs in newly planted seedlings. Amount and plant availability of soil water is dictated largely by the size distribution of individual soil particles. Under non-saturated conditions, coarse-textured soils have greater soil porosity and a lower bulk density resulting in a lower water holding capacity, while finer-textured soils have a greater portion of smaller pores and thus hold more water (Hillel 1971). At soil water contents below saturation, coarse-textured soils can also have a decrease in hydraulic conductivity which reduces water movement to roots (Kramer 1983). Consequently, coarse textured soils, compared to finer-textured soils, have lower water storage capacity, which causes rapid drying from the upper portions of the soil; a lower hydraulic conductivity of coarse textured soils also causes slower replenishment of the soil with water from other regions of the soil profile. As a result, the resistance for water movement through the soil and water uptake by seedlings increases with coarser textured soils under non-saturated conditions (Dosskey and Ballard 1980; Örländer and Due 1986). This can cause seedlings planted into coarse textured soils to undergo planting stress. For example, Engelmann spruce seedlings planted in coarse textured soils had increased seedling water stress compared to seedlings planted in finer-textured soils (Grossnickle and Reid 1984) which resulted in increased mortality (Grossnickle and Reid 1982). In this example successful establishment was affected by hydraulic properties of soil within the rooting zone. Root-zone soil water deficits, especially in coarser textured soils, can be a major cause of planting stress in newly planted seedlings.

Competing vegetation can capture a large portion of the available water, creating conditions where water stress limits seedling performance (Newton and Comeau 1990). Root competition occurs whenever roots of different individual plants are sufficiently close to one another to modify the root environment to the detriment of either or both individuals (Sutton and Tinus 1983). As a result, soil water availability decreases as plant cover increases (Adams et al. 1991).

Root growth is reduced when seedlings are exposed to soils with low water content. This occurs as root growth is reduced directly by water deficits through its effect on cell turgor and cell division (Hsaio 1973). Roots of conifers

and hardwoods cease elongation and become inactive at a soil water potential below -0.06 to -0.07 MPa (Kaufmann 1968; Leshem 1970; Day and MacGillivray 1975; Larson 1980). Thus, planting stress is exacerbated when seedlings are planted into dry soils. Seedlings under water stress are unable to develop new roots needed to acquire sufficient soil water to relieve water stress and resume growth.

Soil temperature

Distinct soil types have differing abilities to transfer heat throughout the soil profile. Organic soils, compared to mineral soils, have poor heat transfer capabilities to underlying soil layers with the organic surface layer acting as an effective barrier in limiting heat penetration down into the soil profile (Geiger 1980; Stathers and Spittlehouse 1990). Thus, there is a reduction in growing season root-zone soil temperatures on sites with an organic layer overlying mineral soil (Balisky and Burton 1997). When energy from the sun is not transferred to the underlying soil layers it is retained in the surface organic soil layer, with surface temperatures of these soils typically 10 – 15 °C higher than mineral soils (Geiger 1980). The result is that soils with an organic surface layer can have high surface temperatures but low root zone temperatures.

Vegetation cover can affect soil temperatures. Removal of vegetation cover often causes an increase in growing season field site soil temperatures (McMinn 1982; Grossnickle and Heikurinen 1989; Stathers and Spittlehouse 1990; Örlander et al. 1990; Coates et al. 1991; Hoog and Lieffers 1991; Wood and von Althen 1993; Burgess et al. 1995; Reynolds et al. 1997; Groot et al. 1997). In these trials, the removal of vegetation cover consistently increased soil temperature by 2 – 5 °C across the growing season. This slight, but continual increase in soil temperature can have a cumulative effect on the growth of conifer seedlings (Margolis and Brand 1990).

Root growth is reduced by low soil temperatures. For example, root elongation of major gymnosperm and angiosperm taiga tree species decreased from 1.5 to 2.0 mm day⁻¹ to a low of 0.5 mm day⁻¹ as soil temperature (@ 10 cm) decreased from 12 to 2 °C (Tryon and Chapin 1983). These findings are consistent with other studies showing that low root temperatures reduce root growth of coniferous species (Stone and Schubert 1959; Lyr and Hoffman 1967; Nambiar et al. 1979; Grossnickle and Reid 1983; Lopushinsky and Kaufmann 1984; Grossnickle and Blake 1985; Lopushinsky and Max 1990; Sutton 1991; Simpson and Ritchie 1997; Ryypö et al. 1998; Landhäusser et al. 2001). Thus, low soil temperature is an edaphic condition that can reduce root growth and subsequent seedling establishment.

Atmospheric demand for water

Atmospheric humidity has a strong effect on the hydrologic cycle of a forest ecosystem through its effect on evapotranspiration. Humidity of the air around a plant is critical because it affects rates of seedling transpiration, which in turn have a direct influence on water movement through seedlings, and on their

nutrient uptake and gas exchange processes. The vapor pressure deficit (VPD – difference between the saturation water vapor pressure of air and ambient vapor pressure of unsaturated air at a given temperature) can be viewed as an indicator of the drying power of air. Vapor pressure deficit changes in an interdependent fashion with changes in both air temperature and relative humidity. The VPD of air is the driving force that cycles water back into the atmosphere within the hydrologic cycle through evaporation of water from soil and transpiration from plants. As VPD increases the water potential (Ψ) of seedlings declines (Ritchie and Hinckley 1975; Hinckley et al. 1978). When seedling Ψ declines seedling growth slows because root growth is reduced directly by seedling water deficits (see above discussion). For example, root growth of black spruce (*Picea mariana* (Mill.) B.S.P.) seedlings declines when grown under high VPD conditions (Darlington et al. 1997). Increasing evaporative demand can create conditions where seedling water stress reaches a level that can cause reduced root growth.

Planting stress does not occur when newly planted seedlings have abundant soil water and low atmospheric evaporative demand. Under these conditions, minimal new root growth is required because the existing root system is adequate to supply water to the shoot system to meet transpirational demand (Simpson and Ritchie 1997). However, it is under these low stress conditions that seedlings readily develop roots into the surrounding soil. This is why soil water content and atmospheric evaporative demand have a direct influence on whether newly planted seedlings undergo water stress and ultimately whether these environmental variables, along with soil temperature, affect the capability of newly planted seedlings to develop roots.

Seedling water balance and planting stress

Soil-plant-air continuum

Water moves along a pathway from the soil through the plant and then into the air through the soil-plant-atmosphere continuum (SPAC). The driving force for this water movement along the SPAC is a water potential (Ψ) gradient between soil, plant and the atmosphere which is in turn driven by the vapor pressure gradient between leaves and the air. As stomata open in early morning and transpiration occurs, water starts to move through the SPAC from a source of higher Ψ (i.e., the soil) through the plant into the air under the pull of atmospheric evaporative demand (i.e., VPD). This pattern of water flow through the SPAC system is analogous to the flow of electricity in a conducting system and can therefore be described by the Ohm's Law analogy (Hinckley et al. 1978; Pallardy et al. 1995) where:

$$\text{Water Flow} = \text{difference in } \Psi / \text{resistance to water and vapor flow}$$

As water flows along this pathway, any condition limiting water flow (i.e., higher resistance) along the SPAC decreases Ψ in plants. The result is that

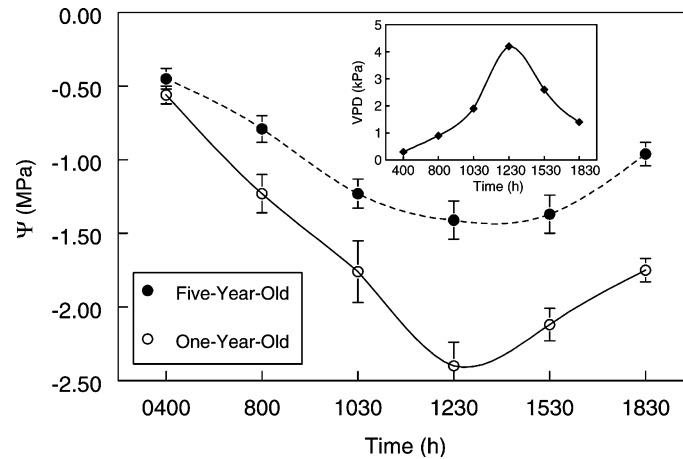


Figure 3. Mid-summer diurnal water status (shoot water potential – Ψ) of newly planted and established container-grown lodgepole pine seedlings planted on a reforestation site in Colorado (Grossnickle 1983). Insert figure represents the vapor pressure deficit (VPD) measured during the day.

plant Ψ normally changes throughout the day in a diurnal fashion (Figure 3). Typically, established seedlings have a Ψ that reaches a daytime low of between -0.75 and -2.00 MPa depending upon field site environmental conditions (Grossnickle 2000). Seedlings are often exposed to a daytime Ψ that can cause water deficits that limit some physiological processes (Hsaio 1973) without causing damage. Stomatal closure and cuticular resistance prevent excess water loss from the plant. Stomata, then, play a critical role in regulating water flow and maintaining Ψ in a seedling at levels such that physiological processes are not damaged.

Newly planted seedlings can be stressed when their water status reaches very low midday minimum Ψ (Figure 3). In these instances of planting stress, young transplanted seedlings can have low photosynthetic rates (Stupendick and Shepherd 1980; Guehl et al. 1989; Grossnickle 1993), which result in the low accumulation of photosynthates (Guehl et al. 1993). This depletion of photosynthates can result in the loss of root regenerating ability in seedlings (Ritchie 1982; Puttonen 1986). Thus, water stress in transplanted seedlings can limit root growth by a lack of water and photosynthates; this can lead to an inability to regenerate roots that are required for establishment.

At night, water moves through the SPAC from a source of higher Ψ (i.e., the soil) to a lower Ψ in the shoot system of seedlings. Since stomata are usually closed at night, water uptake is sufficient for the seedling and the soil to reach comparable values of Ψ . Seedling Ψ just prior to sunrise (Figure 3) (i.e., pre-dawn shoot water – Ψ_{pd}) reflects a plants equilibration to the root zone soil water potential and determines the water status at which a plant begins a daylight period (Ritchie and Hinckley 1975).

Soil and root resistance

As water moves from soil through the seedling and into the atmosphere, one of the primary points of resistance to its flow is at the soil–root interface as water enters the seedling root system (Hellkvist et al. 1974; Running and Reid 1980). Therefore, edaphic and seedling factors limit water flow and create the potential for planting stress (Figure 1). The following section describes how newly planted seedlings respond to edaphic factors of temperature and water. The following section also describes how newly planted seedlings respond to a suberized root system, proper root/soil contact and root confinement. These plant factors are important because seedling water uptake initially depends on the nursery cultured root system for water uptake, though successful establishment depends on root development to avoid long term root confinement and meet the water needs of an expanding shoot system (Carlson and Miller 1990).

Edaphic factors. Plants can have restricted water uptake in cold soils even if there is adequate water within the soil profile (Kramer 1942). This is because there is an increase in plant resistance to water flow with colder soil temperatures that corresponds to an increase in seedling water stress in many conifer species (Kaufmann 1975; Running and Reid 1980; Grossnickle and Blake 1985; Carlson and Miller 1990; Day et al. 1990). White spruce seedlings with an actively growing root system show an increase in seedling resistance which parallels the change in water viscosity, while seedlings with no new roots have an even greater increase in seedling resistance as root temperature declines (Figure 4). This response is attributed to the increased viscosity of water and/or reduced root permeability (Kaufmann 1975). The availability of soil water is one of the most critical environmental parameters required for successful seedling establishment. Under drought, soil hydraulic conductivity is reduced and the rate of water movement through the soil profile to the seedling root system is restricted (see above). Figure 5 shows how white spruce seedlings have increased resistance to water flow through the SPAC with lower soil water availability (i.e., lower Ψ_{pd}). As relative resistance increases under lower soil water availability seedlings come under greater water stress (Kaufmann 1979) because their root systems are unable to take up sufficient water from the surrounding soil to meet the evaporative demands placed upon the shoot system by the atmosphere (Elfving et al. 1972; Hinckley and Buckerhoff 1975; Sterne et al. 1977).

Root suberization. Root resistance to water uptake is variable, depending upon the amount of new roots a seedling develops just after planting. New unsuberized roots permit a more efficient uptake of water (i.e., lower resistance to water flow) than suberized roots (Chung and Kramer 1975; Sands et al. 1982; Carlson 1986; Grossnickle 1988a; Häussling et al. 1988; Colombo and Asselstine 1989; Rüdinger et al. 1994). For example, root resistance decreased in yellow-cedar (*Chamaecyparis nootkatensis* (D. Don) Spach) seedlings with

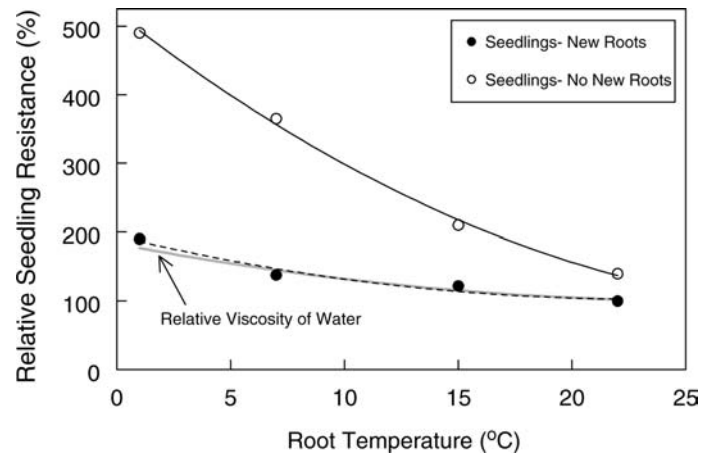


Figure 4. The effect of temperature on relative viscosity of water (thick solid line – scale for viscosity of water runs from 100 to 200% on the y-axis) and the relationship between root temperature and the relative resistance to water flow through the SPAC pathway for bareroot white spruce seedlings with (dashed line and filled symbol) and without (thin solid line and open symbol) new root development (i.e., relative to seedlings with white roots measured at 22 °C) (adapted from Grossnickle 1988a).

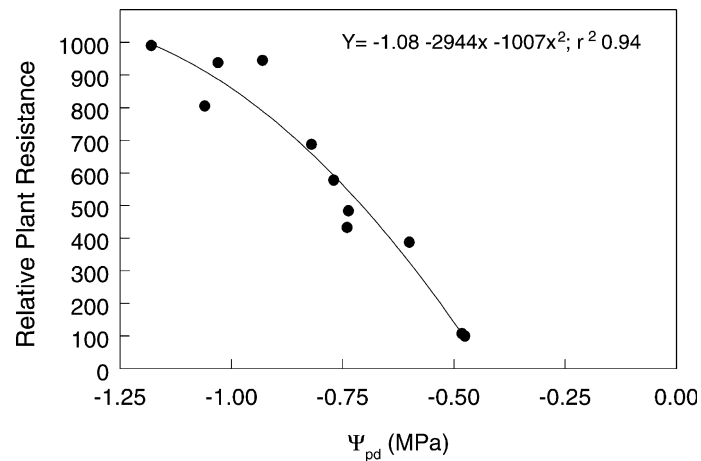


Figure 5. The relationship between seedling predawn water potential (Ψ_{pd}) and the relative resistance to water flow through the SPAC pathway for bareroot white spruce seedlings (i.e., relative to seedlings measured at Ψ_{pd} of -0.45 MPa). (Grossnickle unpublished data).

greater new root development (Figure 6). This ability of newly developed roots to have high water uptake capability allows seedlings to maintain a proper water balance throughout the daytime, thereby decreasing the chance of water stress in recently planted seedlings (Nambiar et al. 1979).

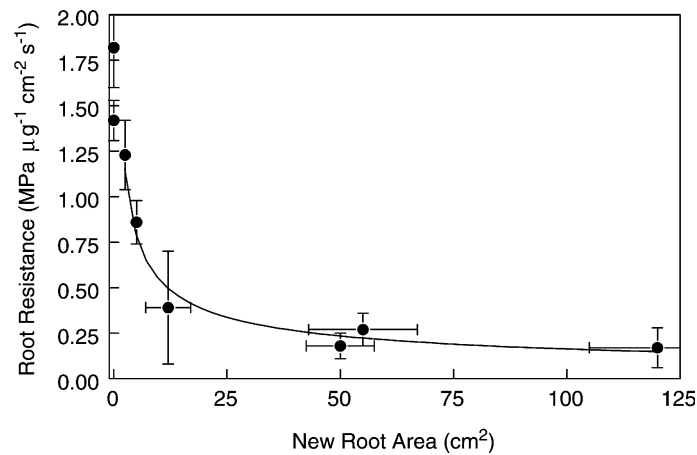


Figure 6. Resistance to water movement root resistance (Root resistance = the slope of the Ψ / Transpiration relationship; Hinckley et al. 1978) through the root system of yellow-cedar seedlings in relation to the amount of new root area (adapted from Grossnickle and Russell 1990).

Root–soil contact. During the planting process, soil is placed around the seedling root system and firmly packed in by the planter. Due to the irregular nature of soil particles, air gaps may occur at the root–soil interface. These air gaps can result in planting stress (Sands 1984), as water can cross this air gap only in the vapor phase, which is considerably less efficient than liquid water movement (Kramer 1983). These air gaps result in a greater resistance to water flow from the soil to the root system of a newly planted seedling. For example, bareroot jack pine seedlings that were planted into soil of a reforestation site had 25–35% greater plant resistance to water movement compared to similar seedlings that were placed into aerated water buckets on the same site (Figure 7). In another example, about half of the plant resistance to water movement in recently planted shining gum (*Eucalyptus nitens* (Deanne and Maiden) Maiden) was attributed to poor root–soil contact (Wilson and Clark 1998). These trials indicate that the lack of sufficient root–soil contact can cause an initial increase in plant resistance to water movement in recently planted seedlings. Stirzaker and Passioura (1996) further confirmed that the main reason for high resistance to water movement in recently planted seedlings is the failure of freshly disturbed soil to pack properly around roots. Wilson and Clark (1998) found that improved root–soil contact caused a decline in plant resistance to water movement over the first 10 days after planting. This was the result of soil particles starting to settle and pack around the root system and the initiation of new root growth.

Root confinement. The lack of root development (i.e., root confinement) can result in increased seedling water stress (Baldwin and Barney 1976; Sands 1984; Draper et al. 1985; Kaushal and Aussenac 1989; Bernier 1993). Newly planted

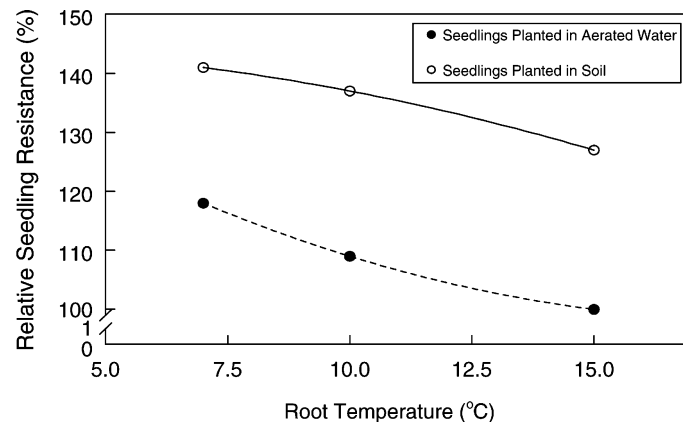


Figure 7. The effect of root-soil contact on the relative resistance to water flow through the SPAC for bareroot jack pine seedlings 1 day after being planted in soil or aerated water on a reforestation site at various soil temperatures (i.e., relative to seedlings planted in aerated water at 15 °C). Seedlings measured had just come out of cold storage, had no new root development and were planted across the range of soil temperatures typically found during a spring planting program (adapted from Grossnickle 1988a).

seedlings have a higher resistance to water movement through the SPAC than older seedlings that have root systems developed out into the surrounding soil (Figure 8). This results in greater water stress for newly planted seedlings

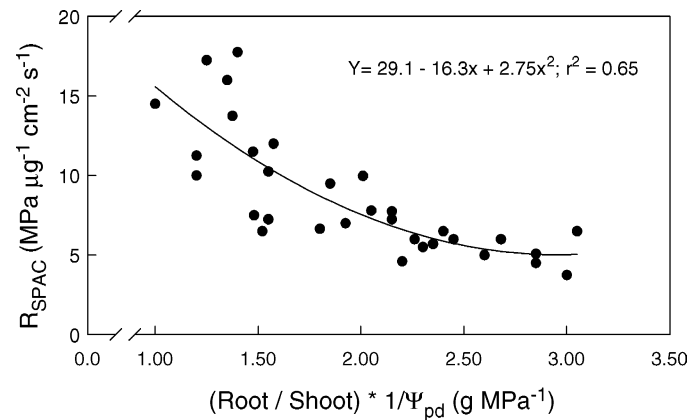


Figure 8. The effect of root growth on resistance to water flow through the SPAC (SPAC resistance = the slope of the $\Psi_{\text{plant}} - \Psi_{\text{soil}}$ /Transpiration relationship; Hinckley et al. 1978) for container-grown Engelmann spruce seedlings planted on a reforestation site over a number of growing seasons. Seedling resistance is related to their total root dry weight/total shoot dry weight $\times$ 1/predawn shoot water potential (i.e., seedling water absorbing system/seedling transpiring system $\times$ available soil water). Recently planted seedlings have a lower ratio while established seedlings have a higher ratio. Each point represents an average of three seedlings (Grossnickle 1983).

compared to older seedlings with well-developed root systems (Figure 3). Little root development into the surrounding soil limits newly planted seedlings access to soil water and results in water stress.

Root growth and seedling water status

A normal pattern of plant Ψ returns as root growth occurs in newly planted seedlings. For example, recently planted container-grown lodgepole pine seedlings had lower minimum daily water potential ($\Psi_{\min}$) during the first two months after planting than seedlings growing for 5 years on the same reforestation site (Figure 9), reflecting minimal root development during the first half of the growing season (Grossnickle and Reid 1983). Towards the end of the growing season $\Psi_{\min}$ of recently planted lodgepole pine seedlings was comparable to seedlings growing for 5 years on the reforestation site. Recently planted seedlings had developed sufficient roots to allow for adequate water uptake from the soil. This phenomenon of improved seedling water status with new root growth has been reported for newly planted seedlings (Nambiar et al. 1979; Grossnickle and Reid 1984; Nambiar 1984; Grossnickle 1988a; Grossnickle and Heikurinen 1989; Omi et al. 1991; Brissette and Chambers 1992; Girard et al. 1997; Jacobs et al. 2004). As seedlings grow roots out from their point of planting into the soil they have access to more soil water and, as a result, have a better seedling water balance (Burdett 1990; Carlson and Miller 1990; Margolis and Brand 1990; Grossnickle 2000). New root growth leads to an increase in daily seedling Ψ , except under limiting environmental conditions, and seedling physiological processes (e.g., photosynthesis, nutrient uptake) begin to resume normal functionality (Sands 1984; Guehl et al. 1989;

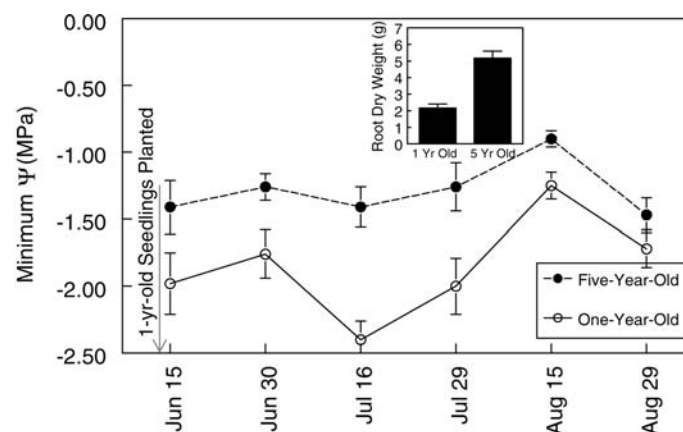


Figure 9. Water status (minimum daytime shoot water potential; mean $\pm$ SE) of newly planted (1-year-old) and established (5-year-old) lodgepole pine container-grown seedlings planted on a site in Colorado (Grossnickle 1983). Insert figure represents the amount of seedling root growth (mean $\pm$ SE) measured at the end of August.

Carlson and Miller 1990; Brisette and Chambers 1992; Grossnickle 1993; Wilson and Clark 1998). The potential for damaging water stress levels is reduced with new root development, thereby improving seedling establishment after planting.

Stocktype performance

Throughout the world most reforestation programs use either bareroot or container-grown stocktypes. In areas where extensive comparison trials have been conducted between these two stocktypes across a range of sites, there is no clear consensus favoring survivability of one particular stocktype (reviewed by Hobbs 1984; Owston 1990; Grossnickle 2000). However, in a number of trials field survival is better for container-grown seedlings when planted on droughty sites (Arnott 1975; Hobbs and Wearstler 1983; Burdett et al. 1984; Nilsson and Örlander 1995). Increased survival of container-grown seedlings was related to lower plant water stress (Hobbs and Wearstler 1983; Nilsson and Örlander 1995). One reason reduced water stress can occur just after planting is that container-grown seedlings can have lower resistance to water flow through the SPAC, compared to bareroot seedlings (Dixon et al. 1983; Grossnickle and Blake 1987). Container-grown seedlings can have greater root growth than bareroot seedlings during the first field growing season (Burdett et al. 1984). Burdett and co-workers (1984) felt this improved root growth minimized water stress and thus relieved the planting stress phenomenon in container-grown, compared to bareroot, seedlings exposed to stressful field site conditions.

Established seedlings

Seedlings enter the establishment phase when they have developed a root system that couples them to available soil water of the field site. Even though seedlings are now established, there can be a carry-over effect from the phase when seedlings did not have well developed root systems. The following examples demonstrate how the planting stress phase can affect subsequent seedling performance even after seedlings have started to develop roots into the soil.

Seedling nutrient status

Recently planted seedlings can have a lower nutrient status than established seedlings. This occurs because seedling nutrient status and subsequent growth are tied to the internal mobilization of nutrients to sites of active growth and the external uptake of nutrients from the soil (Nambiar and Fife 1991; van den Driessche 1991). Seedlings take up nutrients from the soil through a number of mechanisms. First, nutrients are available to seedlings through a passive uptake process. As roots grow into the soil, they come in contact with exchangeable nutrients. Second, nutrients sometimes move towards roots in a soil solution along a concentration gradient through the process of diffusion.

Third, as seedling transpiration occurs there is a mass flow of ions as water flows through the SPAC along the water potential gradient.

Newly planted seedlings, which have very little root development and limited water uptake capability, can have lower nutrient status than established seedlings during the first growing season (Munson and Bernier 1993; Kim et al. 1999). Stored nutrients must be the major source of nutrient reserves for initial seedling development because newly planted seedlings cannot exploit the surrounding soil until they develop new roots (van den Driessche 1991). Limited access to site nutrient reserves by newly planted seedlings is why, in certain instances, foliar nutrient concentrations have been reported to decline during the first growing season (Burdett 1990). Planting seedlings with high initial nutrient reserves (Timmer 1997) or the use of slow release fertilizers placed near the root system (van den Driessche 1991) can improve their growth in the field. Improved seedling nutrient status in the field ultimately occurs when planted seedlings are able to access soil nutrient sources as root systems develop out into the surrounding soil (Munson and Bernier 1993; Imo and Timmer 2001). Thus, adequate root growth is required for improved seedling nutrient status.

Growth check

Growth check, or planting check, is a phenomenon that occurs in conifer seedlings and has long-term implications on plantation performance. Growth check can be defined as reduced shoot growth in a planted seedling that is below the expected shoot growth for an established seedling of equivalent size (South and Zwolinski 1997). Restricted shoot growth often occurs in recently planted spruce seedlings (Armson 1958; Mullin 1963, 1964; Eis 1966; Stiell 1976; Vyse 1981; Burdett et al. 1984; South and Mason 1991; Sutton 1992) and pine seedlings (Hunt and Gilmore 1967; Bacon et al. 1977; Tuttle et al. 1987; Larsen et al. 1988; Zwolinski et al. 1994; South and Smith 2000). Growth check usually starts just after planting and normally lasts for 1–3 years. Seedlings need to avoid long term growth check because established seedlings normally can grow at a rate needed to stay ahead of competing vegetation that begins to reinvade the reforestation site.

Growth check has been attributed to a number of physiological mechanisms, including limited fertility (Sutton 1975) and water stress (Grossnickle 1988a, b). Burdett et al. (1984) proposed that there are two phases to growth check: (1) a newly planted seedling's initial development is altered by water stress, and (2) continued limitations to seedling growth are caused by restricted access to site nutrients. Growth check is overcome when planted seedlings have root growth which allows for a high plant water status and nutrient uptake. As the effects of planting stress disappear seedling shoot growth is similar to that of non-transplanted seedlings of comparable morphology (South and Zwolinski 1997).

Conclusion

Some degree of stress is unavoidable under most planting conditions. Thus, stress can be minimized by addressing the following issues during the planting process. Planting stress can be mitigated somewhat by planting seedlings with high stress resistance and good root growth capability. Planting stress can be minimized by preparing favorable planting sites and planting seedlings properly. Using proper planting practices quickly reduces the lack of proper root-soil contact, minimizes seedling water stress and allows the seedling to initiate new root growth. Lastly, planting stress can be reduced by timing planting to limit exposure to stressful environmental conditions that reduce the physiological response and root growth of seedlings.

Seedlings become established when they have developed a root system that couples them to available soil water of the field site. Proper silviculture practices in combination with good quality seedlings ensure that recently planted seedlings have a chance to develop roots out into the surrounding soil system. Thus, root growth and seedling establishment are tied together when foresters talk about successful reforestation programs.

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